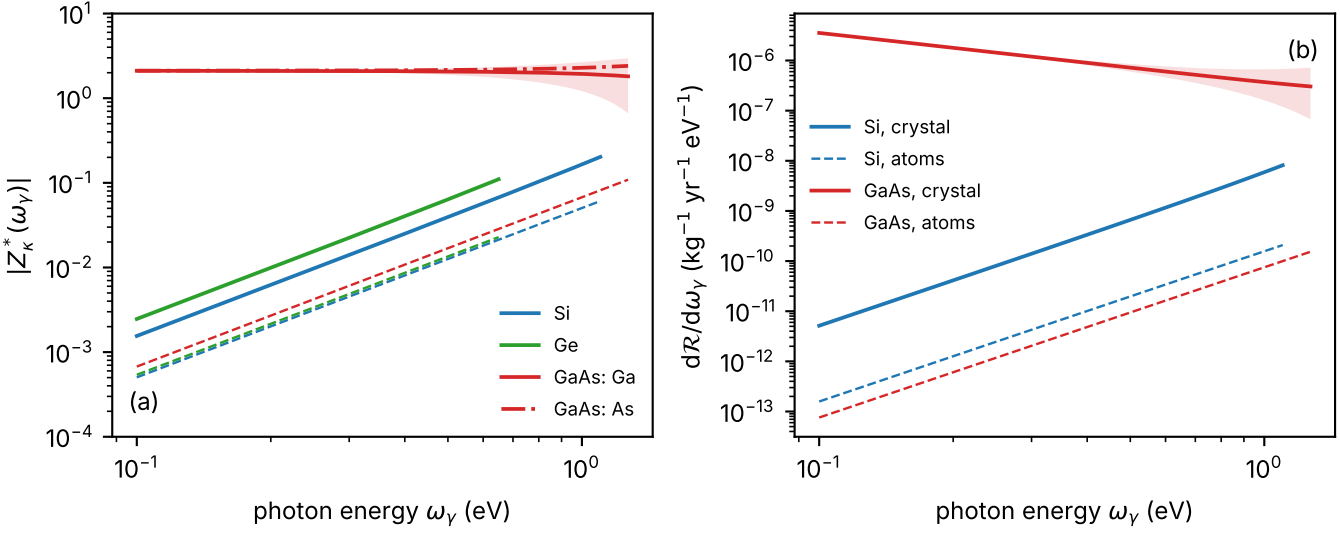
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
$|Z^*_\kappa(\omega_\gamma)|$
Si
Ge
GaAs: Ga
GaAs: As
(a)
10^{-1}
10^0
photon energy ω_γ (eV)
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
$d\mathcal{R}/d\omega_\gamma$ (kg^{-1} yr^{-1} eV^{-1})
(b)
Si, crystal
Si, atoms
GaAs, crystal
GaAs, atoms
10^{-1}
10^0
photon energy ω_γ (eV)